ENGINE TUNE-UP

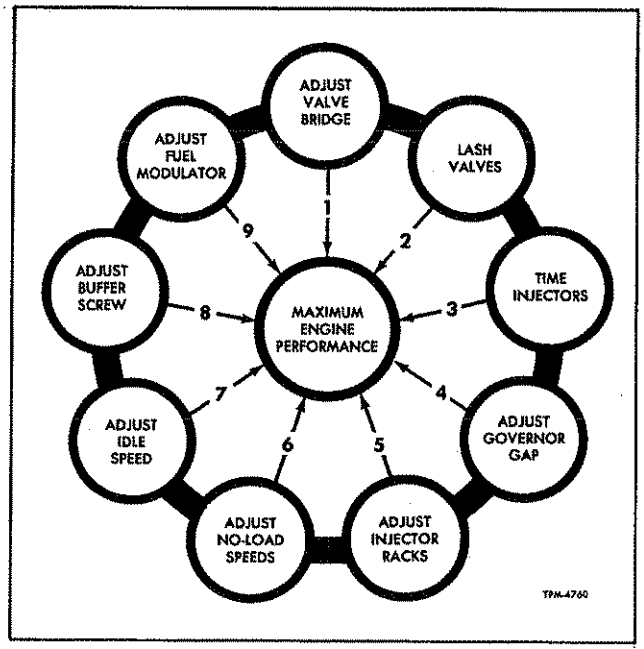


Figure 1—Engine Tune-up Chart

TUNE-UP CHART

Results obtained from an engine tune-up may be unsatisfactory if a "hit and miss" method is used instead of a systematic approach to the job; therefore the logical solution is a complete checkup following the cycle in figure 1 and carrying out each step as directed below.

NOTE: Before tune-up procedure is started it is important that air cleaners and fuel filters are serviced as described in applicable maintenance manuals. Crankcase breather tube and air box drains must be clean and unobstructed. Air box drains may be cleaned with compressed air.

CAUTION: Remove or at least loosen an air box hand hole cover, otherwise blower or end plate gaskets may be damaged by excessive air pressure.

Reference should be made to "TESTING AND DIAGNOSIS" also to "TROUBLE SHOOTING" for engine diagnosis, trouble symptoms, causes, and remedial measures.

The following tune-up adjustments are applicable to two and four-valve cylinder head engines.

TUNE-UP PROCEDURES

To completely tune-up an engine, all adjustments except bridge balancing adjustment and exhaust valve cold setting must be performed only after engine has reached its normal operating temperature.

Results obtained from an engine tune-up are usually unsatisfactory, unless a step-by-step, systematic, and orderly approach is used. Refer to figure 6 and proceed in the following sequence:

1. Bridge Balancing Adjustment (4 Valve Head)
2. Exhaust Valve Lash Adjustment
3. Time Fuel Injectors
4. Adjust Governor Gap
5. Adjust Injector Control Racks
6. Adjust Maximum No-Load Speeds
7. Adjust Idle Speed
8. Adjust Buffer Screw
9. Adjust Fuel Modulator (Turbo-Charged)

NOTE: Whenever a push rod has been disconnected from the clevis, the push rod must be screwed back into place until end of push rod appears through the clevis. If this is not done, the piston may hit the head of the valve when the engine is being turned, due to the small clearance, between the valves and piston head at the piston's upper position, or an injector plunger may "Bottom" before being properly adjusted.

Clearance between valve rocker arm and exhaust valve stem, or bridge with engine at operating temperature is important and should be maintained. Too little clearance causes a loss of compression, misfiring of cylinders and eventual burning of valves and valve seats. Too much clearance results in noisy operation of the engine, especially in the low speed range.

BRIDGE BALANCING ADJUSTMENT

The exhaust valve bridge assembly is adjusted and the adjustment screw locked securely at the time the cylinder head is installed on the engine. Until wear occurs with the operation of the engine, no further adjustment is required on the exhaust valve bridge. When wear is evident, make complete bridge balancing adjustment as outlined in the following steps: